PROMPT ENGINEER BUNDLE

PROMPT ENGINEERING – Basics

1. INTRODUCTION

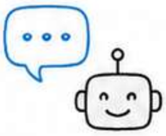
Prompt Engineering is the art and science of crafting input (prompts) to get accurate, relevant, and useful responses from AI language models.

Why it matters?

- Improves response quality
- Saves time and reduces trial & error
- Helps AI understand context better
- Enables consistent and reliable outputs

KEY TOPICS

1. What is a Prompt?



A prompt is the input or instruction given to an AI model to guide its response.

Example:

"Explain photosynthesis in simple words."

2. Clarity and Specificity

Clear and specific prompts produce better results. Vague prompts lead to general or off-topic answers.

Example:

✗ "Tell me about trees."

✓ "Explain the uses of mangrove trees in coastal ecosystems."

3. Context is Important

Providing context helps the model understand the background and give more relevant responses.

Example:

With context: "As a marketing manager, write a short email to announce a new product launch."

4. Role Prompting

Assigning a role to the AI helps frame the response in the desired style, tone, or perspective.

Example:

"You are a helpful travel guide. Suggest a 3-day itinerary for Paris."

Quick Tip:

Good prompts = Clear instructions + Relevant context + Desired output format

PROMPT ENGINEERING NOTES

1 Zero-Shot Prompting

- Ask the model to perform a task without providing any examples.

Example:

Prompt: Explain the benefits of regular exercise.

Use when:
The task is simple and the model is already familiar with it.

2 One-Shot Prompting

- Provide one example to show the model what kind of output is expected.

Example:

Prompt: Q: What is the capital of France?
A: Paris

Q: What is the capital of Japan?
A:

Use when:
You want to guide the model with a single example.

3 Few-Shot Prompting

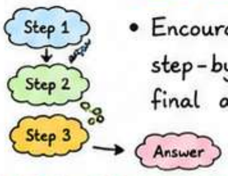
- Provide a few examples to show the model what kind of output is expected.

Example:

Prompt: Q: Translate 'Hello' to Spanish.
A: Hola
Q: Translate 'Thank you' to Spanish.
A: Gracias
Q: Translate 'Good morning' to Spanish.
A:

Use when:
The task is complex or the model needs more guidance on the format or style.

4 Chain-of-Thought (CoT) Prompting



- Encourage the model to think step-by-step before giving the final answer.

Example:

Prompt: If a notebook costs $3 and a pen costs $2, how much will 4 notebooks and 3 pens cost? Let's think step by step.

Use when:
The task involves reasoning, math, planning, or multi-step problem solving.

5 Role Prompting

- Assign a specific role or persona to the model to get responses in a particular tone or style.

Example:

Prompt: You are a financial advisor. Suggest 3 tips for saving money for college students.

Use when:
You want the response in a specific tone, style, or from an expert perspective.

Quick Tip: The more clearly you guide the model with the right technique, the more accurate, relevant, and useful the response.

PROMPT ENGINEERING NOTES

Advanced Techniques

1. Delimiters

Use delimiters to clearly separate different parts of the prompt such as instructions, context or examples.

Example:

Context: <context> ... </context>
Instruction: <instruction> ... </instruction>
Input: <input> ... </input>

Helps the model understand the structure of your prompt.

2. Output Format Control

Specify the desired format of the response like bullets, table, JSON, or a specific template.

Example:

Prompt: List 3 benefits of meditation.
Format: Respond in a JSON array with fields "benefit" and "description".

Helps get responses that are easy to parse and use in applications.

3. Constraints and Rules

Set limits or rules to guide the model on what to include or avoid in the response.

Example:

Prompt: Write a summary of this article.
Constraints: Use less than 100 words. Do not add opinions. Stick to the facts.

Constraints reduce irrelevant or unwanted outputs.

4. Tone and Style Control

Instruct the model to respond in a specific tone, style, or audience level.

Example:

Prompt: Explain photosynthesis.
Tone & Style: Explain like you're teaching a 10-year-old in a friendly and simple way.

Helps match the response with your audience and use case.

5. Iterative Prompting

Refine the prompt based on the model's output. Improve it step by step to get better results.

Example:

Step 1: Ask the question.
Step 2: Review the output.
Step 3: Refine the prompt and ask again.

Small changes in prompts can lead to much better responses.

6. Combining Techniques

Combine multiple techniques like context, examples, format and constraints for best results.

Example:

Use role + examples + format + constraints together to build powerful prompts.

The right combination makes your prompts highly effective.

Key Takeaway:

Great prompts are clear, structured and tailored. Use the right technique for the right task, and keep improving!

PROMPT ENGINEERING NOTES

Improving Prompts

1 Use Clear and Specific Language

Clear prompts reduce ambiguity and get better responses.

Example:
- ✗ "Tell me about AI.
- ✓ "Explain the applications of AI in healthcare with real-world examples."

Tip: Be specific about what you want.

2 Provide Context

Give background or context so the model understands better.

Example:
- ✗ Write a report.
- ✓ Write a report on the impact of remote work on productivity.

Tip: Context = Better relevance.

3 Ask for Structured Output

Request the output in a specific format.

Example:
Prompt: List 5 popular programming languages.
Better Prompt: List 5 popular programming languages in a table with columns: Language, Year of Release, Use Case.

Tip: Specify format like table, list, JSON, steps, etc.

4 Use Examples

Examples guide the model to understand the pattern and style you want.

Example:
Prompt: Classify the sentiment of the following review as Positive, Negative or Neutral.
Review: "The product is amazing and works well."
Sentiment: Positive
Review: "Not what I expected. Waste of money."
Sentiment: Negative

Tip: Examples = Better accuracy.

5 Iterate and Refine

Refine the prompt based on the response to improve results.

Example:
Step 1: Ask the question.
Step 2: Review the response.
Step 3: Refine the prompt with more details.
Step 4: Repeat until satisfied.

Tip: Small changes can make a big difference.

BEST PRACTICES

- Be specific about the task.
- Provide enough context.
- Specify the output format.
- Define the audience or use case.
- Review and refine your prompts regularly.
- Combine techniques for best results.

COMMON MISTAKES TO AVOID

- ✗ **Vague Prompts** — Being unclear leads to generic answers.
- ✗ **Too Many Requests** — Asking too much at once can confuse the model.
- ✗ **Irrelevant Details** — Extra information that doesn't matter can distract the model.
- ✗ **Ignoring Output Format** — Output ? Not specifying format may give unstructured responses.
- ✗ **Not Iterating** — Not refining prompts means missing better results.

Remember: Great prompts = Clear intent + Relevant context + Right techniques + Continuous improvement.

PROMPT ENGINEERING NOTES

1. Re-Act (Reason + Act)

A prompting framework where the model alternates between reasoning and acting using tools or environments.

Example Workflow:

Thought: I need today's weather.
Action: Search["weather in Paris"]
Observation: Sunny, 22°C
Thought: Now I can answer.
Answer: It's sunny and 22°C in Paris.

2. Prompt Template

A reusable structure with placeholders to ensure consistency and scalability.

Example Template:

You are a {role}.
Your task is to {task}.
Use the following context:
{context}.
Answer in {format}.

3. User Prompt

The input or query provided by the user to get a response.

Example:

Explain the benefits of regular exercise for productivity.

Clear and specific user prompts lead to better responses.

4. System Prompt

Initial instructions that set the behavior, persona, and rules for the model.

Example:

You are a helpful assistant.
Answer truthfully, be concise,
and avoid harmful content.

The system prompt defines the model's role and boundaries.

5. Temperature

Controls randomness in the output.
Higher values = more creative,
lower values = more focused.

Example:

0.0 —— 0.7 —— 1.0

Use 0.2 – 0.4 for factual answers.
Use 0.7 – 1.0 for creative writing.

6. Top-p (Nucleus Sampling)

Controls diversity by sampling from the smallest set of tokens whose cumulative probability ≥ p.

Example:

0.0 —— 0.9 —— 1.0

Use 0.8 – 0.95 for balanced results.
Lower = more focused, Higher = more diverse.

7. Top-k Sampling

Limits the model to the top k most likely next tokens.

Example:

1 —— 50 —— 200

Lower k = more focused.
Higher k = more diverse.

Quick Reminders

- Start with clear instructions (System Prompt).
- Provide enough context.
- Tune parameters (Temperature, Top-p, Top-k) based on the task.
- Evaluate and iterate for better results.

8. Prompt Evaluation Metrics

Relevance
How well the response addresses the prompt.
Measure: Human score, embedding similarity

Coherence
Logical flow and consistency of the response.
Measure: Human score

Fluency
Grammatical correctness and readability.
Measure: Perplexity, Human score

Faithfulness
Factual accuracy and consistency with the given context.
Measure: Fact-check, Human score

Completeness
Whether the response covers all aspects of the prompt.
Measure: Human score

Helpfulness
Overall usefulness and actionability of the response.
Measure: Human score

Conciseness
Whether the response is concise and to the point.
Measure: Length, Human score

Creativity
Originality and inventiveness of the response.
Measure: Human score

Safety
Absence of harmful, biased, or inappropriate content.
Measure: Safety filters, Human score

Great Prompts + Right Settings + Evaluation = Powerful AI Responses!

PROMPT ENGINEERING NOTES

7 Persona Prompting

Assign a persona or expert role to the model.

Example:
Prompt: You are a nutritionist. Suggest a healthy vegetarian meal plan for a week.

Personas guide tone, style, and depth of response.

8 Step-back Prompting

Ask the model to think broadly before going deep.

Example:
Prompt: Before solving this math problem, explain the key concepts involved.

Helps in complex reasoning and better understanding.

9 Self-Consistency

Ask the model to generate multiple answers and choose the best one.

Example:
Prompt: Solve this puzzle in three different ways and choose the most reliable answer.

Improves accuracy on reasoning and math problems.

10 Tree of Thoughts (ToT)

Explore multiple solution paths like a tree and choose the best.

Example:
Prompt: Solve this riddle by exploring different possible approaches and pick the best solution.

Useful for complex problems with many possible solutions.

11 Generated Knowledge Prompting

Ask the model to generate knowledge or facts before using them.

Example:
Prompt: First list key facts about climate change. Then explain how it affects coastal cities.

Builds better, fact-based and informed responses.

12 Automatic Reasoning

Let the model reason step-by-step without manual instructions.

Example:
Prompt: If a train travels 120 km in 2 hours, how far will it travel in 5 hours? Show your reasoning.

Encourages natural reasoning and logical thinking.

13 Hypothetical Prompting

Ask "what if" questions to explore possibilities.

Example:
Prompt: What if AI could read human emotions? How would it impact society?

Great for creative thinking and exploring scenarios.

14 Contrastive Prompting

Ask the model to compare two or more options.

Example:
Prompt: Compare renewable and non-renewable energy sources in terms of cost, impact, and future potential.

Helps in decision-making and understanding trade-offs.

15 Multi-Modal Prompting

Use text, image, or data together for better results.

Example:
Prompt: Describe this image and suggest a caption for social media.

Combines different inputs for richer and relevant responses.

16 Reflection Prompting

Ask the model to review its own response and improve it.

Example:
Prompt: Review your answer above. Can you make it more clear and concise?

Improves clarity, correctness, and overall quality.

EXTRA TIPS

- Keep prompts short, clear, and focused.
- Use examples to explain your intent.
- Avoid giving too much information.
- Test different phrasings to get better results.
- Iterate: small changes can make a big impact!

PROMPT FORMULA

Role / Persona + Task / Instruction + Context + Format + Constraints + Tone = Great Response!

Great prompts are a mix of Art and Science. Keep practicing, experimenting, and improving!